Second Batch Community Experiment — Week #2 Problems

First Proof

July 1, 2026

1 Problem C3

Fix a compact Lie group $\mathbf{G}$ with Lie algebra $\mathfrak{g}$, endowed with an Ad-invariant norm. We define invariant $\mathcal{C}^{-\kappa}_{\mathrm{inv}}$ "norms" on smooth $\mathbf{G}$-equivariant connections B_0 on the trivial $\mathbf{G}$-bundle over the torus $\mathbb{T}^2$ by

$$\|B_0\|_{\mathrm{inv},-\kappa} = \sup_{s\in(0,1]} s^{\frac{\kappa}{2}} \Big(s^{\frac{1}{2}} \|F(s)\|_{L^\infty} + s\|\mathrm{D}F(s)\|_{L^\infty} \Big) \,. \tag{1}$$

Here $F(s) = F_{B(s)}$ denotes the curvature tensor for the solution $B(s)$ at time s to the Yang–Mills gradient flow with initial condition B_0 and $\mathrm{D}F(s)$ denotes the covariant derivative of $F(s)$ with respect to the connection $B(s)$. The quantity given by (1) is of course not a genuine norm since the relation $B_0 \mapsto F$ is nonlinear. Recall also that the group of smooth functions $g\colon \mathbb{T}^2 \to \mathbf{G}$ acts on such connections B_0 by

$$g \bullet B_0(x) = \mathrm{Ad}_{g(x)} B_0(x) - dg(x)\, g(x)^{-1} \,.$$

Give a proof of the following theorem.

Theorem 1. *Given $\kappa \in (0, \frac{1}{2})$, there exist constants $c, C > 0$ such that the following is true. For every smooth connection B_0 as above, there exists a gauge transformation g such that*

$$\|g \bullet B_0\|_{\mathcal{C}^{-\kappa}} + \sup_{s\in(0,1]} s^{\kappa} \|\mathrm{div}(g \bullet B(s))\|_{L^\infty} \le C(1 + \|B_0\|_{\mathrm{inv},-\kappa})^c \,. \tag{2}$$

2 Problem C4

Give a proof of the following theorem.

Theorem 2. *Given $c > 1$ and $G_n \sim G(n, c/n)$, let C_{max} denote the giant component. Conditional on $x, y, z \in C_{max}$, sample $[x, y], [y, z]$ and $[z, x]$ uniformly at random (from the set of shortest paths between the corresponding vertex pairs) to form $\Delta(x, y, z)$.*

For every fixed $\epsilon > 0$, there exists a constant $M(\epsilon) > 0$ such that for the scale

$$\delta_n := \left\lfloor \frac{1}{\log c} \left(\frac{1}{2} \log n - 9 \log \log n \right) \right\rfloor - M(\epsilon),$$

there exists N such that for all $n \geq N$, conditioned on the uniformly randomly chosen vertices x, y, z lie in the giant component of G_n,

$$\mathbb{P}_{G_n}(\mathbb{P}(\mathrm{slim}(\Delta(x, y, z)) > \delta_n) > 1 - \epsilon) > 1 - \epsilon$$

where $\Delta(x, y, z)$ is a geodesic triangle.